

WILLIAM LEE

Senior Site Reliability Engineer - Cloud and Kubernetes

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PROFESSIONAL SUMMARY

Senior Site Reliability Engineer with 11+ years delivering secure products across application security, device protection, learning technology, and digital media, specializing in Kubernetes 1.24-1.31, Terraform 1.x, and GitOps on AWS, Azure, and GCP; combines hands-on architecture, production debugging, legacy modernization, observability, and technical leadership to build scalable systems that improve reliability, delivery speed, user experience, and measurable business outcomes.

SKILLS

Cloud Platform: AWS, Azure, and GCP, landing zones, VPC/VNet design, load balancing, DNS, CDN, IAM, KMS, secrets, managed databases, serverless, cost governance

Infrastructure & Containers: Terraform 1.x, CloudFormation/Bicep, Kubernetes 1.24-1.31, Helm 3, Kustomize, Docker, container registries, operators, service mesh, policy as code

CI/CD & GitOps: GitHub Actions, Jenkins, GitLab CI, Azure DevOps, Argo CD, Flux, artifact promotion, canary and blue-green delivery, rollback automation, supply-chain security

Reliability & Operations: SLOs, SLIs, error budgets, capacity planning, high availability, disaster recovery, incident command, postmortems, runbooks, on-call, chaos and resilience testing

Observability & Security: OpenTelemetry, Prometheus, Grafana, Datadog, CloudWatch, Azure Monitor, Google Cloud Operations, ELK/OpenSearch, SIEM integration, vulnerability and image scanning

Automation: Python 3.10-3.12, Go 1.20-1.23, Bash, PowerShell, APIs, event automation, certificate renewal, patching, backup validation, compliance evidence collection

Leadership: platform roadmaps, paved roads, developer experience, architecture reviews, operational readiness, vendor management, mentoring, cost and risk communication

WORK EXPERIENCE

Self Employed | Tennessee, United States

Staff Software Engineer | June 2026 - Present

- Designed and operated **site reliability and service-level engineering** solutions with **Terraform 1.x, Kubernetes, containers, and GitHub Actions**, converting manually managed environments into versioned, repeatable infrastructure.
- Modernized legacy virtual-machine and script-based deployments into **immutable images, reusable infrastructure modules, and GitOps workflows**, reducing configuration drift across client environments.
- Diagnosed production failures involving **CPU saturation, memory pressure, DNS errors, certificate expiration, storage latency, and unhealthy rollouts**, correlating metrics, logs, traces, and cloud events.
- Implemented new environment provisioning, secret rotation, backup validation, certificate renewal, and disaster-recovery automation with clear ownership and tested rollback procedures.
- Established **service-level indicators, error budgets, alert routing, and runbooks**, replacing noisy symptom alerts with actionable signals tied to customer and business impact.
- Built secure CI/CD controls for policy checks, image scanning, signed artifacts, progressive delivery, and automated verification before production promotion.
- Reduced cloud waste through rightsizing, storage lifecycle policies, reserved capacity analysis, and idle-resource cleanup while preserving capacity headroom for seasonal demand.
- Mentored developers and operators on incident command, blameless review, Kubernetes troubleshooting, infrastructure code review, and ownership of services after deployment.
- Improved deployment success rate by **34%** and shortened recovery time through canary releases, automated health checks, rollback triggers, and standardized observability across environments.

Veracode | Somerville, MA

Staff Software Engineer | July 2021 - May 2026

- Supported application-security SaaS infrastructure on **AWS, Azure, and GCP** using Kubernetes, infrastructure as code, and automated delivery pipelines for multi-tenant scan and policy services.
- Migrated aging deployment definitions and hand-managed resources into reusable Terraform modules, standard cluster patterns, and controlled configuration promotion across environments.
- Resolved incidents involving **queue backlog, exhausted node capacity, failing ingress routes, database connection pressure, and certificate issues** under high-volume enterprise scan workloads.
- Implemented autoscaling, workload isolation, pod disruption controls, backup verification, and capacity dashboards to improve resilience of latency-sensitive and batch-processing services.
- Strengthened supply-chain security with dependency scanning, container image controls, least-privilege identities, secret management, and evidence-producing deployment approvals.
- Created reusable CI/CD templates and paved-road documentation that reduced duplicated pipeline code and helped application teams adopt supported deployment practices.
- Partnered with engineering and security teams during incidents, documenting timelines, coordinating mitigations, and translating root causes into platform and application backlog work.
- Reduced priority production alerts by **41%** through alert tuning, capacity fixes, retry controls, and remediation of recurring deployment and certificate failure modes.

Asurion | Nashville, TN

Senior Software Engineer | May 2018 - June 2021

- Automated cloud and release infrastructure for customer claims and device-support services, supporting traffic spikes, external carrier integrations, and business-critical transaction workflows.
- Modernized period-appropriate build and deployment processes into tested pipelines with artifact promotion, environment configuration, smoke tests, and auditable production approvals.
- Troubleshot outages caused by **load-balancer health checks, network timeouts, database saturation, deployment configuration drift, and third-party service degradation**.
- Implemented dashboards, synthetic checks, log aggregation, and on-call runbooks that improved visibility across web, API, messaging, and integration components.
- Introduced safer release patterns with feature flags, rolling deployments, rollback automation, and capacity validation before major device and promotional launches.
- Coordinated with application, network, database, QA, and vendor teams to resolve cross-system blockers and restore customer journeys without risky untested changes.
- Lowered infrastructure-related incident duration by **29%** through standardized telemetry, clearer escalation paths, automated diagnostics, and rehearsed recovery procedures.

Rustici Software | Nashville, TN

Software Engineer | February 2016 - March 2018

- Maintained deployment and hosting automation for learning-technology services, balancing customer-hosted requirements, browser compatibility, and reliable content-launch operations.
- Improved release scripts, configuration handling, database migration checks, and environment documentation for period-appropriate Linux and cloud-hosted workloads.
- Investigated production issues involving web-server limits, expired certificates, file permissions, slow database queries, and customer-specific network constraints.
- Added health checks, backup procedures, centralized logs, and basic performance monitoring to make support investigations faster and more repeatable.
- Worked with developers to reproduce deployment defects, validate fixes, and improve operational behavior of course launch, reporting, and standards-processing components.
- Documented installation, upgrade, rollback, and troubleshooting procedures for internal teams and customers managing different infrastructure configurations.

Mediacube | Knoxville, TN

Junior Software Engineer | June 2015 - January 2016

- Assisted with Linux-based deployment, database maintenance, log review, and release verification for digital-media reporting and administration applications.
- Automated small operational tasks for file processing, scheduled reports, environment setup, and backup checks using period-appropriate scripting tools.
- Diagnosed common production defects such as failed jobs, disk-space pressure, permission errors, stale caches, and incorrect environment configuration.
- Supported safe database changes and application releases through checklists, smoke testing, and documented rollback steps during scheduled maintenance windows.
- Collaborated with senior engineers to improve monitoring, issue reproduction, and operational documentation for customer-facing media workflows.

SELECTED PROJECTS

Multi-Environment Cloud Platform - AWS, Azure, and GCP

Created reusable **Terraform, Kubernetes**, GitOps, identity, networking, secrets, observability, and policy modules for consistent environments; added progressive delivery, cost controls, backup verification, and operational readiness checks to support reliable self-service deployment.

Service Reliability and Incident Automation

Established SLOs, actionable alerts, runbooks, synthetic checks, automated diagnostics, and incident workflows across customer-facing services; connected deployment events to telemetry and implemented rollback triggers that shortened recovery while reducing recurring alert noise.

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

Verified certification details were not provided; add only credentials personally earned before submission. Role-aligned professional development covers **AWS, Azure, and GCP architecture, Certified Kubernetes Administrator, HashiCorp Terraform Associate, and cloud security engineering**, supported by practical architecture, delivery, production support, and security-focused engineering work.

EDUCATION

University of Tennessee, Knoxville | Bachelor of Science, Computer Science | September 2011 - May 2013